Kushagra Bhatore

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Education

National Institute of Technology, Jamshedpur — Jamshedpur, India

*B.Tech. in Production and Industrial Engineering (CGPA: **8.38/10 across 6 semesters**)* — *Aug 2023 – Present*

- **Relevant Coursework:** Introduction to Programming & Data Structures, Basic Electrical Engineering, Engineering Mathematics I & II (Linear & Vector Algebra, Calculus), Operations Research, Industrial Automation & Control Systems, Probability & Statistics.

Technical Skills

Languages & Core CS: Python, C/C++, TypeScript, SQL, Data Structures & Algorithms (DSA), Object-Oriented Design (OOD), DBMS, Operating Systems

Backend & Frontend: FastAPI, Node.js, REST APIs, WebSockets, React, Next.js, HTML5, CSS3, Tailwind CSS

Databases & Cloud: PostgreSQL, Redis, SQLite, Databricks, Delta Lake, Microsoft Azure, AWS

Data & ML Libraries: PySpark, pandas, NumPy, scikit-learn, XGBoost

Dev Tools, Testing & AI: Git, GitHub, Linux CLI, Docker, pytest, Hypothesis, fast-check, Claude Code, GitHub Copilot, Cursor

Work Experience

Nielsen India Pvt. Ltd. — Bangalore, India

Software & Automation Engineering Intern — *Jun 2026 – Aug 2026*

- Engineered Python-based financial automation services and backend workflows, integrating automated tool-calling and validation rules to resolve reporting anomalies across enterprise systems.
- Architected a centralized backend gateway via **OData**, exposing **32 RESTful APIs** with dynamic authentication, observability logging, and strict rate-limiting to replace manual cross-team data requests.
- Automated end-to-end data reconciliation and natural-language query execution across **48 relational tables** (**50M+ records**), cutting end-to-end batch execution and reporting latency from hours to **under 4 minutes**.

Projects

Convergence – Real-Time Collaborative Sync Platform **[Live]** — Feb 2026 – May 2026

TypeScript, Node.js, Rust (napi-rs), Next.js, WebSockets, PostgreSQL, Redis, Vitest, fast-check

- Built a full-stack real-time collaboration platform using **LWW-Element-Set CRDTs**, Hybrid Logical Clocks, and a native **Rust (`napi-rs`)** merge addon achieving **50K–206K+ ops/sec** throughput at **sub-5ms P50 latency**.
- Designed a channel-multiplexed **WebSocket** server (**1–5ms batch windows**) supporting isolated multi-room sessions, **60fps** live cursor presence, and automatic state snapshots with garbage collection every **1,000 ops**.
- Developed a local-first **TypeScript** client SDK featuring a **100K-op offline queue**, exponential-backoff reconnection, time-travel state replay, and interactive Next.js demos (collaborative whiteboard, split-screen, stress-test suite).
- Verified CRDT convergence, causal ordering, and room isolation across **926 automated tests** (**643 backend, 283 frontend** in Vitest) including **43 property-based tests** (**fast-check**).

Real-Time Network Telemetry & Anomaly Detection Platform — Aug 2025 – Dec 2025

Python, FastAPI, asyncio, SQLite, XGBoost, scikit-learn, pytest, Hypothesis

- Engineered an asynchronous telemetry ingestion backend in Python (**asyncio, FastAPI**) processing **50+ network flows/sec** with **<1ms local overhead**, streaming live alerts via Server-Sent Events (SSE).
- Implemented an in-memory pub/sub event bus using bounded asynchronous queues with backpressure handling and **SQLite (WAL mode)** persistence to prevent worker starvation under traffic spikes.
- Integrated a **12-class XGBoost classifier** (**0.91 precision**) alongside an asynchronous IP aggregation engine correlating live traffic against **60,000+ malicious IPs** from 4 external threat feeds.
- Enforced system reliability across queue bounds, rate-limiting, and database persistence via **270+ unit and property-based tests** (**pytest, Hypothesis**).

Achievements & Leadership

- **Algorithmic Problem Solving & DSA:** Solved **1000+ Data Structures and Algorithms problems** across competitive programming platforms, achieving peak ratings of **1535 on LeetCode** and **1313 (Pupil) on Codeforces**.
- **Smart India Hackathon (SIH) National Finalist:** Led end-to-end system architecture, backend API development, and live prototype execution for a national-level problem statement among top engineering teams in India.
- **National Winner – Zerodha Varsity Quiz:** Secured **1st place nationwide** across premier engineering colleges.
- **Core Organizer – Qunite (College Quizzing Club):** Directed operations and hosted flagship inter-college quizzes for **500+ participants**.